An Analysis of Systemic Bottlenecks in Mathematics Performance: A Case Study of a Girls' Secondary School in Machakos County

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Introduction and Background

A student's performance in Mathematics significantly shapes their trajectory in STEM-related careers and their future competitiveness in the global labor market. Beyond individual success, proficiency in Mathematics is a key metric of institutional prestige. As a woman in STEM, I am committed to bridging the gender gap in technical subjects. This commitment led me to undertake a pro bono consulting project for my alma mater—a C2-level girls' high school in Machakos County—where Mathematics has historically been the lowest-performing subject.

Problem

Despite continuous institutional investments in teacher-led remedial programs and strategic planning by the school board, mathematics performance at the institution has remained critically low and stagnant. Traditional interventions have historically treated performance drops as a pedagogical issue, assuming that more teaching hours or modified instructional styles would rectify the deficit. However, initial trend analysis indicates a cohort-wide decline that triggers immediately after the first term of Form 1, neutralizing even the strongest academic foundations (high KCPE entry marks). The persistent failure of standard remedies suggests a fundamental misdiagnosis of the problem.

There is a critical lack of empirical data regarding non-pedagogical variables - specifically student mindsets, behavioral sub-habits, and classroom-level dynamics - and how these factors interact to undermine academic execution. Without isolating the true systemic drivers of this decline, the institution risks wasting limited resources on non-performing solutions while the students face long-term exclusion from competitive STEM career paths.

Methods

Design

I utilized an exploratory research design to investigate the non-pedagogical drivers of mathematics performance. The study operationalized variables across two primary domains to capture a holistic view of the learning environment:

- **Student-Related Factors:** Measured through sub-domains of Interest, Attitude, and Study Habits.
- **Teacher-Related Factors:** Measured through sub-domains of Teaching Style, Expectations & Support, and Communication & Availability.

Each domain was assessed using a structured 15-item instrument utilizing a 5-point Likert scale (ranging from Strong Disagreement as 1 to Strong Agreement as 5) to quantify student perceptions and behavioral frequencies.

Sampling and Data Ethics

The target population comprised the Form 3 cohort, totaling 250 students distributed across five distinct streams. To ensure unbiased representation across varying classroom environments, a probability-based random sampling technique was employed, yielding a final sample size of n = 56 students.

To protect data integrity and mitigate social desirability bias, the instrument was administered completely anonymously. Students were provided explicit assurances of strict confidentiality before administration of the questionnaire to encourage honest and uninhibited reporting.

Data Processing and Analysis

Data processing used a complementary dual-software approach designed to maximize cleaning rigor and statistical depth:

- **Microsoft Excel:** Utilized for initial data entry, structural screening, missing value auditing, and baseline descriptive visualizations.
- **R Language for Statistical Computing:** Utilized to execute advanced inferential modeling, including multiple linear regression analysis and multi-stream interaction modeling to isolate cohort sub-dynamics.

Findings

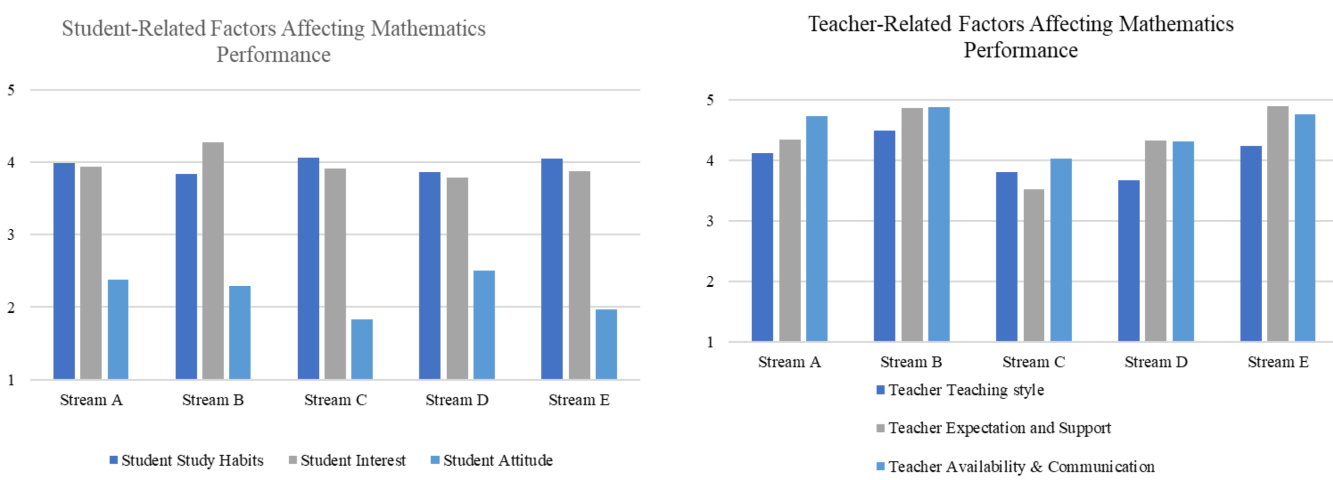
Descriptive Statistics

The descriptive findings (Figure 1) indicated a striking structural contrast between institutional inputs and student baseline dynamics across the cohort. Teacher-Related Factors - encompassing teaching style, expectations, and communication availability – remained generally high across the board, averaging well above the mid-point. However, the classroom environment was not entirely uniform. Streams C and D exhibited noticeable institutional dips, particularly within "Teacher Expectation and Support" and "Teacher Teaching Style," where scores fell below the 4.0 threshold seen in other streams.

Student-related factors indicate a major bottleneck, compared to the teacher domain. While student study habits and general interest levels remained moderately high and consistent across the board, Student Attitude emerged as the primary hindrance to performance across every single stream. Attitude scores were systematically the lowest metric recorded, plunging to a critical low in Stream C (< 2.0) and Stream E (2.0). This is despite the differences in teacher-related scores, such as the high scores in Stream E and the lower scores in C. This uniform deficit points to localized behavioral and/or psychological barriers transcending the classroom environment.

Figure 1.

Comparative Presentation of Institutional Support vs. Student Factors by Stream



Regression Analysis

Table 1

Summary of Multiple Regression Analysis for Variables Predicting Outcome

Variable	B	*t*-value	Pr(>|*t*|)
Intercept	42.25	2.16	0.039*
Student Interest	1.39	0.419	0.677 NS
Student Attitude	-7.53	-3.02	0.0040*
Student Study Habits	7.63	2.01	0.050*
Teacher Teaching Style	-0.98	-0.31	0.75 NS
Teacher Exp & Support	-2.23	-0.079	0.45 NS
Teacher Availability & Comm	-0.34	-0.079	0.93 NS

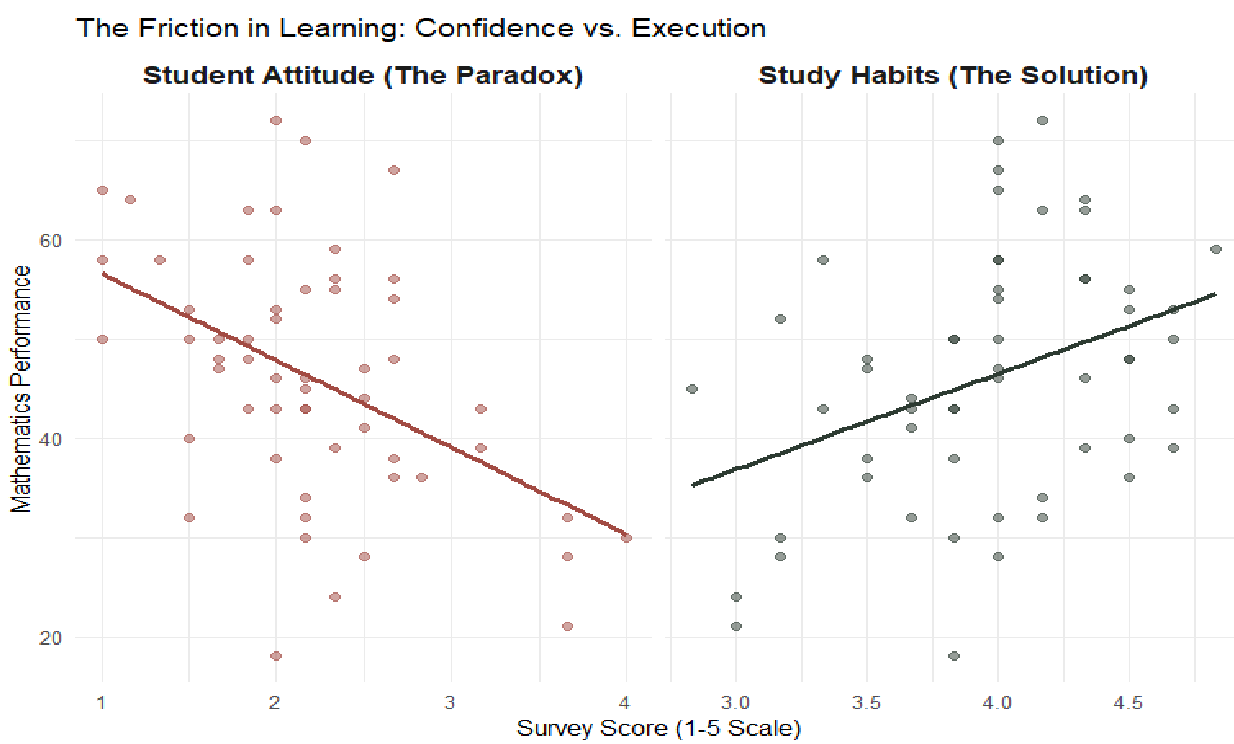
Note. (n = 56). Dependent variable: Performance. $R^2 = 0.29$; Adjusted $R^2 = 0.20$; ($F(6, 49) = 3.33$, p = 007). **B** = Unstandardized regression coefficient. NS = Not Significant; *p ≤ 0.05; **p* < *0.01.*

When interpreting individual predictors, the teacher-related domains did not reach statistical significance, with *p*-values ranging from 0.45 to 0.93. This empirical evidence mathematically demonstrates that variations in teacher performance are not driving the stagnation in grades. Instead, the performance variance is likely more dictated by a behavioral tug-of-war within the students themselves. While Student Study Habits yielded a significant positive effect (B = 7.63, $p = 0.05$), validating traditional revision logic, Student Attitude revealed a profound, counterintuitive penalty (B = -7.53, $p = 0.004$).

Holding all other variables constant, every 1-unit increase in self-reported positive attitude corresponds to a 7.53-point drop in actual test performance. This statistically robust "Overconfidence Paradox" strongly indicates a severe confidence-competence gap. Superficial optimism and a high comfort level with the subject appear to be masking deep academic weaknesses, inadvertently preventing students from engaging in the rigorous, error-correcting revision required to master secondary-level mathematics. Figure 2 provides a visual of how study habits and attitude relate to performance.

Figure 2.

How Student Attitude and Study Habits Affect Performance



To evaluate whether performance variance was driven by specific classroom assignments, a second model was constructed that introduced "Stream" as a categorical control factor (with Stream A as the baseline reference). The addition of the stream variables did not yield a statistically significant improvement in the model's predictive power, and none of the individual stream coefficients were statistically significant. This non-significance proves that no single stream is inherently disadvantaged or failing on its own. A student in Stream B or D doesn't perform worse simply because of the room they sit in or the main teacher assigned to that block. This structural neutrality confirmed that the "Overconfidence Paradox" observed in Model 1 is a cohort-wide psychological phenomenon, rather than a localized environmental failure

Further investigations included testing whether classroom context alters the behavioral dynamics of the cohort, using a multi-stream interaction model. The interaction model ($R^2 = 0.391$, $p = 0.0036$, $F(9,46) = 3.28$, $p = 0.004$) unmasked the localized architecture of the performance bottleneck as indicated in Table 2.

Table 2

Interaction Regression Model for Attitude and Stream Dynamics

Variable	B	*t*-value	Pr(>|*t*|)
Stream A (Baseline/ Intercept)	1.64	0.08	0.939 NS
Student Attitude (Strm A Slope)	17.63	2	0.051
Stream B Main Effect	70.2	3.00	0.004**
Stream C Main Effect	42.93	1.62	0.112 NS
Stream D Main Effect	73.22	3.19	0.003**
Stream E Main Effect	52.88	1.24	0.22 NS
Attitude*Stream B Interaction	-29.67	-3.05	0.004**
Attitude*Stream C Interaction	-15.81	-1.29	0.20 NS
Attitude*Stream D Interaction	-29.79	-3.16	0.003**
Attitude*Stream E Interaction	-19.64	-0.95	0.34 NS

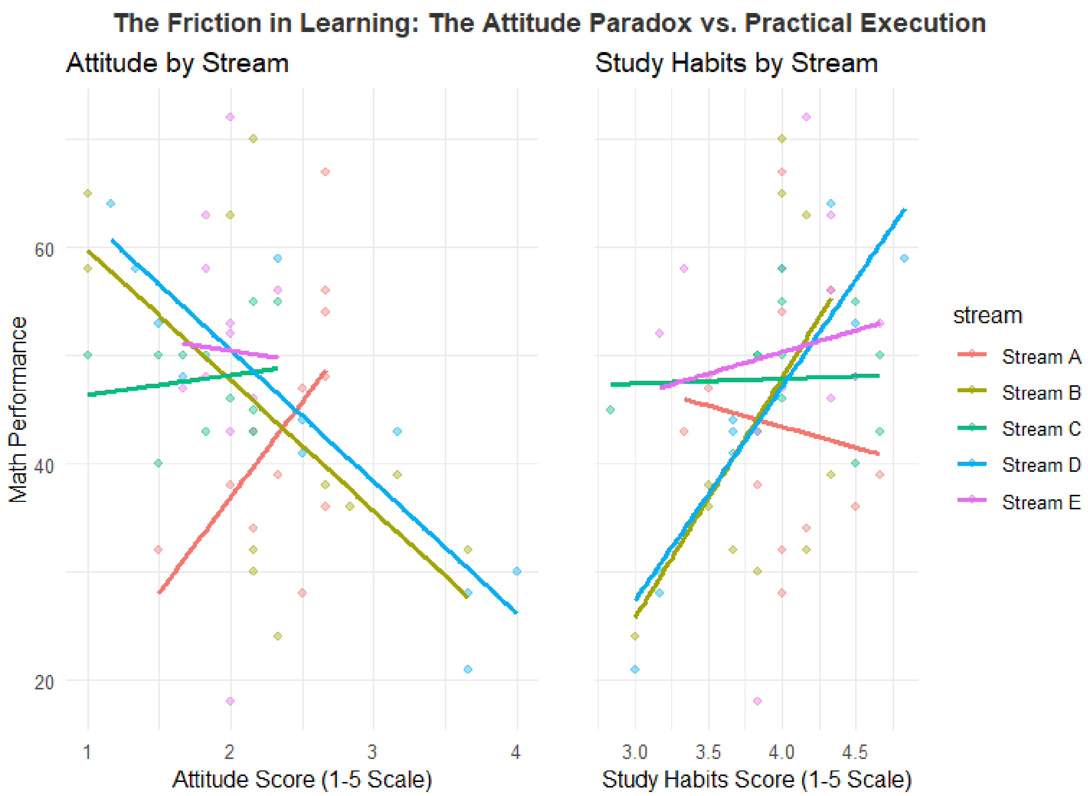
Note. (n = 56). Dependent variable: Performance. Reference group = Stream A. **B** = Unstandardized regression coefficient. NS = Not Significant; **$p \leq 0.005$;

In the reference classroom (Stream A), the relationship between attitude and performance is actually positive ($B = 17.63$, $p = 0.051$). In this environment, student confidence closely aligns with academic execution, meaning optimism acts as a performance catalyst. However, this positive dynamic is entirely reversed and overwhelmed in Streams B and D. The interaction terms for these two specific classrooms are deeply negative and highly significant:

- Stream B Attitude Penalty: B = -29.67, $p = 0.004$(Net effective slope: -12.04)
- Stream D Attitude Penalty: B = -29.79, $p = 0.003$ (Net effective slope: -12.16)

Figure 3

Stream-Wise Variation in the Performance Slopes of Student Attitude and Study Engagement



Holding all other factors constant, an increase in superficial confidence within Streams B and D results in a severe, localized performance crash of over 12 marks. Conversely, the interaction effects for Streams C and E failed to reach statistical significance, indicating a neutral behavioral baseline. This empirical divergence proves that the "Overconfidence Paradox" is not an inherent trait of the students themselves, but rather a product of classroom sub-cultures. In Streams B and D, peer dynamics, a lack of competitive accountability, or specific classroom routines are actively fostering an illusion of competence. Students in these rooms believe they understand the material, probably leading to a premature cessation of deep study, whereas the structural sub-culture in Stream A effectively ties confidence to actual mastery. Figure 3 further illustrates the interaction between attitude, study habits, and stream.

Conclusions and Recommendations

This diagnostic project successfully isolated the systemic undercurrents driving the stagnation of mathematics performance within the Form 3 cohort. The analysis yields three definitive conclusions:

1. **The Myth of the Pedagogical Deficit:** Traditional interventions that focus exclusively on increasing teaching hours or modifying teaching delivery are targeting the wrong variable. Teacher-related factors scored uniformly high across the descriptive analysis and completely failed to reach statistical significance in all inferential models. The institution's teaching staff is a stable, high-performing asset.
2. **The "Overconfidence Paradox":** The primary constraint on performance is a psychological and behavioral mismatch within the student body. Cohort-wide, superficial optimism is heavily penalized. Students are confusing a general comfort level with the material with actual operational mastery, leading to a confidence-competence gap.
3. **Classroom Micro-Cultures:** The interaction model proves that this paradox is not a generic student flaw, but a product of localized classroom environments. While Stream A has structurally aligned confidence with actual execution, Streams B and D possess severe localized sub-cultures where unbacked confidence results in catastrophic performance drops.

Strategic Recommendations

To bridge the confidence-competence gap and dismantle the problematic subcultures in Streams B and D, the institution should pivot from "more teaching" to "rigorous tracking" by implementing the following three low-cost, high-impact strategies:

1. Mandate Objective Feedback Systems (Dethrone "Feeling" Prepared): Given that subjective student attitudes are decoupled from actual performance, students must no longer rely on how they "feel" about a topic.

- **Action:** Introduce mandatory, 10-minute anonymous diagnostic exit tickets at the end of every major mathematical topic.
- **Execution:** These should be binary (Pass/Fail) single-question mini-assessments. If a student who "feels" confident fails the exit ticket, the instant objective feedback shatters the illusion of competence before the end-of-term exams.

2. Institute Weekly Error-Logging Protocols: Study Habits are highly beneficial but are currently being neutralized by poor attitudes.

- **Action:** Replace standard passive revision with active, mandatory Error Logs.
- **Execution:** Students must maintain a dedicated section in their notebooks where they are required to rewrite every problem they got wrong in weekly quizzes, explicitly document the specific logic gap that caused the error, and resolve the problem correctly twice.

3. Cross-Stream Peer Audit (Replicate Stream A's Structure) : Since Stream A is successfully managing to maintain a healthy relationship between attitude and performance, its micro-culture needs to be institutionalized.

- **Action:** The administration should conduct an internal administrative audit of Stream A's daily routines.
- **Execution:** Document how prep-time is monitored, how peer-studying groups are formed, and how questions are asked in Stream A. Pair the class teachers to actively replicate those accountability structures prominent in Stream A.